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CNN–Transformer Encoder–Decoder for Quantitative LIBS Analysis: Explicit Mono–Polyatomic Spectral Decomposition Applied to Volcanic Soil Characterization

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Abstract

Laser-Induced Breakdown Spectroscopy (LIBS) enables rapid multi-element analysis without extensive sample preparation, yet quantifying elemental concentrations in complex geological matrices remains challenging due to spectral overlap, matrix effects, and plasma variability. Existing machine learning models for LIBS treat the measured spectrum as a single composite signal and thus disregard the physically meaningful decomposition into individual elemental (monoatomic) contributions. This work introduces a CNN–Transformer encoder–decoder architecture that explicitly models the relationship between monoatomic component spectra and the observed polyatomic mixture spectrum. The encoder, fed with individual monoatomic spectra produced by a Saha–Boltzmann forward model, learns element-specific latent representations through self-attention. The decoder receives the polyatomic mixture spectrum and recovers elemental concentrations via cross-attention over the encoder output, effectively learning the compositional rules that govern spectral superposition. One-dimensional convolutional layers precede both encoder and decoder to extract local emission-line features before the Transformer captures long-range wavelength dependencies. The model is pre-trained on synthetic spectra generated from NIST Atomic Spectra Database transitions and fine-tuned on experimental LIBS measurements of volcanic soil samples from Aceh, Indonesia, using X-ray fluorescence data as ground-truth concentrations. Baseline comparisons against CNN-only, Transformer-only, partial least-squares, and Informer models are reported. The proposed framework is, to our knowledge, the first to embed an explicit mono–polyatomic spectral decomposition within a deep-learning pipeline for quantitative LIBS analysis.

1 Introduction

Laser-Induced Breakdown Spectroscopy (LIBS) is an emission spectroscopy technique that has been widely applied to elemental analysis of complex materials, including minerals, rocks, and soils [1]. Its ability to perform rapid multi-element analysis with minimal sample preparation makes it attractive for geochemical prospecting and natural-resource exploration [2]. Multiple studies have demonstrated successful elemental quantification in geological samples through approaches ranging from multivariate regression and univariate calibration curves to machine-learning methods [3, 4, 5]. Although these approaches can achieve high analytical accuracy, their performance depends strongly on the spectral modelling strategy and on the sensitivity of the method to matrix effects in compositionally heterogeneous samples [6, 7]. This dependence is particularly pronounced in volcanic soils, which inherently exhibit complex multi-element compositions with large matrix variability. Among quantitative LIBS approaches, Calibration-Free LIBS (CF-LIBS) enables elemental quantification without reference standards by extracting fundamental plasma

parameters, namely the electron temperature T_e and the electron density n_e, under the assumption of Local Thermodynamic Equilibrium (LTE) [8]. However, the assumption of a homogeneous LTE plasma is often inadequate for describing the highly transient, spatially graded plasmas generated by nanosecond laser pulses [9, 10]. Deviations from LTE induce spectral distortions through optical opacity and self-absorption, which destabilise temperature and density estimates [11, 12].

Physics-based approaches such as the MERLIN adaptive radiative-transfer model have addressed these distortions by coupling Saha–Boltzmann population distributions with the radiative transfer equation, thereby reproducing emission spectra with thermodynamic consistency [13, 14]. Despite their physical rigour, these forward models entail computationally expensive nonlinear inversions that limit their applicability for routine analysis. Machine learning and deep learning have emerged as alternatives that learn the nonlinear mapping from spectra to plasma parameters directly from data [14, 15, 16]. Convolutional Neural Networks (CNNs) capture local spectral features such as emission-line profiles, while Transformer architectures model long-range dependencies across the wavelength axis [17, 15]. Hybrid CNN–Transformer models in particular have demonstrated promising results for quantifying trace elements in steel [15] and for classifying spectral signatures in multi-element samples [18]. Nevertheless, most data-driven models operate as black boxes that do not explicitly encode the physics governing spectrum formation [6]. This lack of physical structure limits generalisation and demands large experimental datasets for reliable training. A fundamental physical fact, however, remains unexploited in current modelling practice: a polyatomic LIBS spectrum is, to a first approximation, a concentration-weighted superposition of monoatomic component spectra. All existing models, whether physics-based or data-driven, treat the measured spectrum as a single signal and thereby ignore this compositional structure [15, 14]. No prior work has explicitly modelled the decomposition of a mixture spectrum into its monoatomic constituents within a learning framework. Cross-attention, a mechanism originally proposed for sequence-to-sequence translation [17], is ideally suited for this task: the decoder can query encoder-produced element representations to discover which monoatomic contributions compose a given polyatomic observation.

This paper presents a CNN–Transformer encoder–decoder architecture that introduces explicit mono–polyatomic spectral decomposition into quantitative LIBS analysis. The encoder processes individual monoatomic spectra generated by a Saha–Boltzmann forward model to learn element-specific latent representations. The decoder receives the polyatomic mixture spectrum and predicts elemental concentrations through cross-attention over the encoder output. One-dimensional convolutional layers extract local emission-line features before the Transformer stages capture global wavelength dependencies. The model is pre-trained on synthetic spectra derived from the NIST Atomic Spectra Database and fine-tuned on experimental LIBS measurements of volcanic soil from Aceh, Indonesia, with X-ray fluorescence (XRF) analysis providing ground-truth concentrations. Contributions of this work are threefold: (i) the first encoder–decoder framework that structurally separates monoatomic from polyatomic spectral representations for LIBS, (ii) a cross-attention mechanism that enables physically interpretable concentration regression, and (iii) a validated application to complex volcanic-soil matrices.

2 Methodology

2.1 Framework overview

The proposed approach follows a two-stage workflow. In the first stage *(offline)*, a physics-based forward model generates pairs of monoatomic and polyatomic synthetic spectra across a grid of plasma conditions; these serve as training data. In the second stage *(online)*, a CNN–Transformer encoder–decoder consumes the spectral pairs and predicts elemental concentrations.

2.2 Synthetic data generation

Synthetic spectra are produced under the assumption of LTE, which is reasonable for the early-to-mid temporal window of nanosecond LIBS plasmas [8]. The population of each ionisation stage is governed by the Saha equation [19],

$$\frac{n_e\, n_{z+1}}{n_z} = 2\,\frac{Z_{z+1}(T_e)}{Z_z(T_e)}\left(\frac{2\pi m_e k_B T_e}{h^2}\right)^{3/2} \exp\!\left(-\frac{E_{\mathrm{ion},z}}{k_B T_e}\right), \tag{1}$$

where n_z and n_{z+1} are the number densities of ionisation stages z and $z+1$, Z_z denotes the partition function, $E_{\mathrm{ion},z}$ is the ionisation energy, and the remaining symbols have their usual meanings. Within each ionisation stage the excited-state populations follow the Boltzmann

distribution [19],

$$n_i = \frac{n_z\, g_i \exp(-E_i/k_B T_e)}{Z_z(T_e)}, \tag{2}$$

where g_i and E_i are the degeneracy and energy of level i.

The spectral emission coefficient for a transition $k \to i$ is [19]

$$j_\lambda = \frac{hc}{4\pi\lambda}\, A_{ki}\, n_k\, \phi_V(\lambda), \tag{3}$$

with Einstein coefficient A_{ki} and Voigt profile $\phi_V(\lambda)$. The Voigt profile arises from the convolution of a Gaussian component due to Doppler broadening [20],

$$\Delta\lambda_D = \lambda_{ki}\sqrt{\frac{8\ln 2\, k_B T_i}{m_k c^2}}, \tag{4}$$

and a Lorentzian component dominated by Stark broadening [20],

$$\Delta\lambda_S \approx 2\,\omega_{ki}(T_e)\,\frac{n_e}{n_{e,\mathrm{ref}}}, \tag{5}$$

where ω_{ki} is the electron-impact half-width at reference density $n_{e,\mathrm{ref}}$.

Atomic transition data (λ_{ki}, A_{ki}, g_i, E_i) are taken from the NIST Atomic Spectra Database [21]. The spectral window covers 200–900 nm. Plasma parameters are sampled on a grid with $T_e \in [5\,000,\ 15\,000]$ K and $n_e \in [10^{16},\ 10^{18}]$ cm^{-3}. For each plasma condition and each element z, a monoatomic spectrum $S^{(z)}_{\mathrm{mono}}(\lambda)$ is computed by summing equation (3) over all transitions of that element. The corresponding polyatomic spectrum is constructed as a concentration-weighted superposition,

$$S_{\mathrm{poly}}(\lambda) = \sum_z c_z\, S^{(z)}_{\mathrm{mono}}(\lambda), \tag{6}$$

where c_z is the fractional concentration of element z, with $\sum_z c_z = 1$. Finally, the synthetic spectrum is convolved with an instrumental Gaussian of full width at half maximum $\Delta\lambda_{\mathrm{inst}} \approx 0.1$ nm to match the spectral resolution of the laboratory spectrometer.

2.3 Experimental data

Experimental LIBS spectra are acquired from volcanic-soil samples collected in the Aceh region of Sumatra, Indonesia. The soil pellets are ablated with a Q-switched Nd:YAG laser operating at 1064 nm. Emission is collected and dispersed by an echelle spectrometer coupled to an intensified CCD detector. Each spectrum represents the average of multiple laser shots to improve the signal-to-noise ratio.

Ground-truth elemental concentrations are obtained independently by X-ray fluorescence (XRF) analysis. The XRF concentrations serve as regression targets during fine-tuning, while CF-LIBS derived concentrations [8] provide supplementary validation. Detailed instrumental parameters and sample-preparation protocols are summarised in table 1.

Table 1. Summary of experimental parameters.

Parameter	Value
Laser wavelength	1064 nm
Pulse energy	114 mJ
Gate delay	0.5 μs
Gate width	0.5 μs
Spectral range	200–900 nm
Number of samples	24
Ground-truth method	XRF

2.4 CNN–Transformer encoder–decoder architecture

The architecture comprises four modules: (i) a shared 1D CNN feature extractor, (ii) a Transformer encoder operating on monoatomic representations, (iii) a Transformer decoder with cross-attention operating on the polyatomic spectrum, and (iv) a regression head that outputs elemental concentrations. Figure 1 shows the complete model.

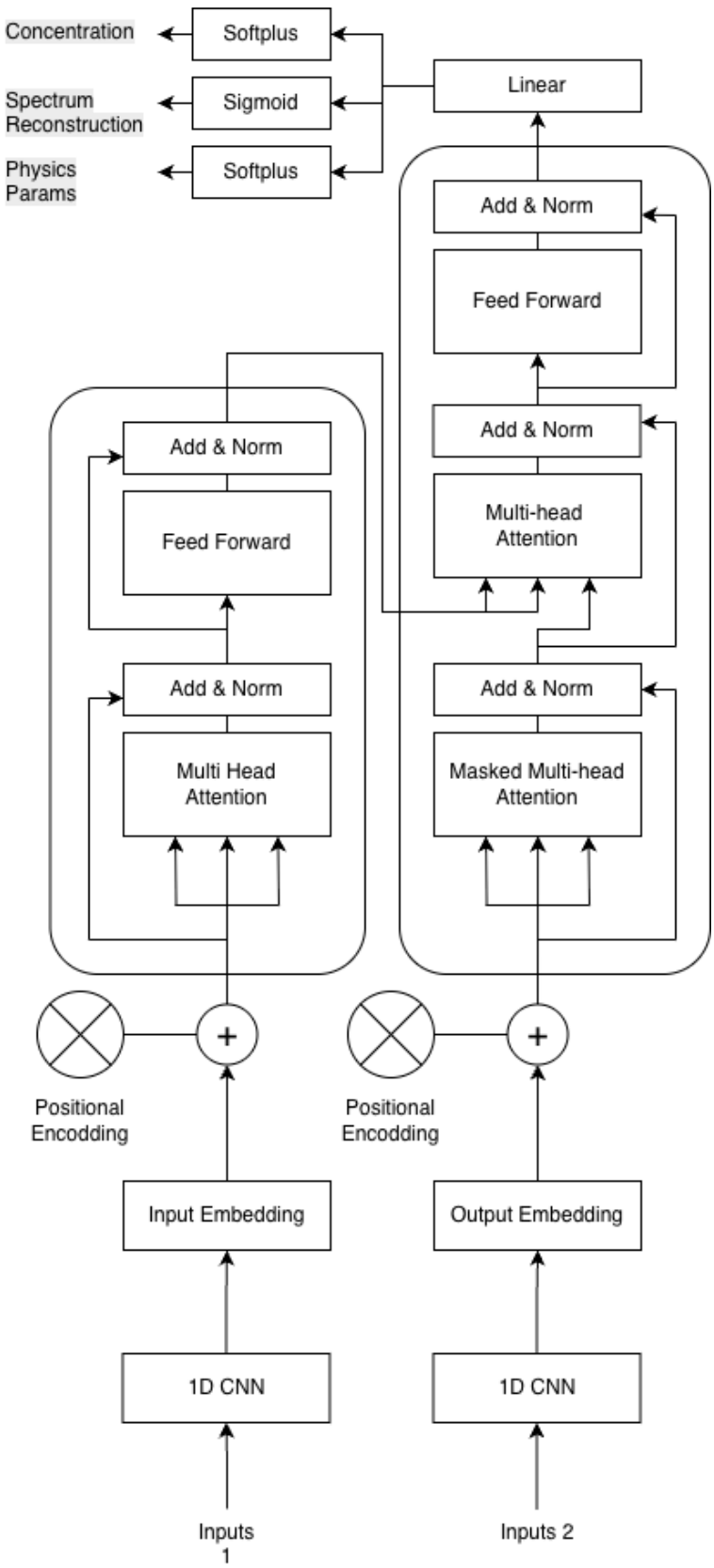

Figure 1. Detailed CNN–Transformer encoder–decoder architecture. The encoder (left) processes monoatomic spectra through 1D CNN, input embedding, positional encoding, and N_{enc} self-attention layers. The decoder (right) processes polyatomic spectra through masked self-attention and cross-attention with the encoder output. Three output heads predict elemental concentrations (Softplus), reconstructed spectra (Sigmoid), and plasma parameters (Softplus).

2.4.1 Input representation. A monoatomic spectrum $S^{(z)}_{\text{mono}}(\lambda) \in \mathbb{R}^L$ is a one-dimensional signal of length L (number of wavelength channels) corresponding to element z. The polyatomic spectrum $S_{\text{poly}}(\lambda) \in \mathbb{R}^L$ is the experimentally measured or synthetically generated mixture signal. Both are normalised to unit maximum intensity before being input to the model.

2.4.2 1D CNN feature extractor. Each spectrum is passed through a stack of 1D convolutional layers [22] that capture local emission-line patterns:

$$\mathbf{F} = \text{ReLU}\big(\text{BN}(\text{Conv1D}(S;\, \Theta_{\text{cnn}}))\big), \tag{7}$$

where BN denotes batch normalisation [23] and Θ_{cnn} are the learnable convolutional parameters. The CNN parameters are shared between the encoder and decoder paths to enforce a common spectral feature space. The output is a sequence of feature vectors $\mathbf{F} \in \mathbb{R}^{L' \times d}$, where L' is the reduced sequence length after pooling and d is the feature dimension.

2.4.3 Transformer encoder. The encoder receives the CNN features of all Z monoatomic spectra, concatenated along the sequence dimension, together with learnable element-type and positional embeddings [17]. Multi-head self-attention learns intra-element and inter-element dependencies:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}. \tag{8}$$

After N_{enc} encoder layers, the output is a set of element-specific latent representations $\mathbf{H}_{\text{enc}} \in \mathbb{R}^{(Z \cdot L') \times d}$.

2.4.4 Transformer decoder with cross-attention. The decoder operates on the CNN features of the polyatomic spectrum [17]. Each decoder layer contains three sub-layers: masked self-attention over the mixture features, cross-attention with the encoder output, and a position-wise feed-forward network. The cross-attention mechanism is the key structural innovation:

$$\mathbf{Q}_{\text{dec}} = \mathbf{F}_{\text{poly}}\,\mathbf{W}_Q, \quad \mathbf{K}_{\text{enc}} = \mathbf{H}_{\text{enc}}\,\mathbf{W}_K, \quad \mathbf{V}_{\text{enc}} = \mathbf{H}_{\text{enc}}\,\mathbf{W}_V, \tag{9}$$

where $\mathbf{F}_{\text{poly}}$ denotes the decoder's polyatomic features and $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ are learnable projections. The resulting cross-attention scores have a physically meaningful interpretation: high attention weight between a decoder position (a spectral region of the mixture) and an encoder position (an element-specific feature) indicates that the corresponding element contributes to the mixture intensity at that wavelength.

After N_{dec} decoder layers, a global average pooling and a linear projection yield the predicted concentration vector:

$$\hat{\mathbf{c}} = \text{Linear}\big(\text{GAP}(\mathbf{H}_{\text{dec}})\big) \in \mathbb{R}^Z. \tag{10}$$

2.4.5 Training strategy. The model is trained in two phases.

Pre-training on synthetic data. The encoder–decoder is trained on $(\{S^{(z)}_{\text{mono}}\}_{z=1}^{Z},\, S_{\text{poly}},\, \mathbf{c})$ triplets generated by the forward model (section 2.2). The loss function combines a concentration regression term and an optional spectral reconstruction term:

$$\mathcal{L} = \frac{1}{Z}\sum_{z=1}^{Z}(\hat{c}_z - c_z)^2 + \alpha\,\|S_{\text{poly}} - \hat{S}_{\text{poly}}\|_2^2, \tag{11}$$

where α is a regularisation weight. The Adam optimiser [24] is used with a learning rate of 10^{-4} and cosine annealing schedule.

Fine-tuning on experimental data. The pre-trained model is fine-tuned on experimental LIBS spectra with XRF-derived concentrations as targets. During fine-tuning the monoatomic spectra are generated by the forward model at plasma conditions estimated from each experimental spectrum. The learning rate is reduced by a factor of ten to preserve the representations learned during pre-training.

Domain adaptation. To further bridge the synthetic–experimental domain gap, a domain-adversarial training strategy is employed. A Gradient Reversal Layer (GRL) is attached to the shared Transformer encoder, feeding into a domain classifier that distinguishes synthetic from experimental spectra. During backpropagation the GRL reverses the gradient sign, encouraging the encoder to learn domain-invariant representations. Figure 2 illustrates the domain-adaptation architecture.

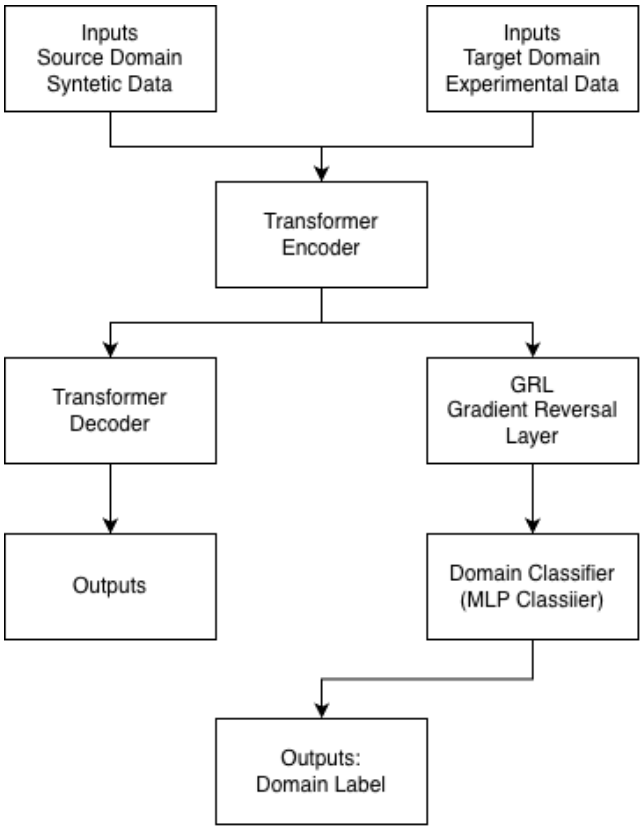

Figure 2. Domain-adaptation architecture. Source-domain (synthetic) and target-domain (experimental) spectra share the same Transformer encoder. A Gradient Reversal Layer (GRL) feeds a domain classifier that encourages domain-invariant encoder representations through adversarial training.

2.5 Evaluation protocol

Model performance is quantified by three metrics computed per element:

- Root Mean Square Error: $\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(c_i - \hat{c}_i)^2}$
- Coefficient of determination: $R^2 = 1 - \frac{\sum(c_i - \hat{c}_i)^2}{\sum(c_i - \bar{c})^2}$
- Mean Absolute Percentage Error: $\mathrm{MAPE} = \frac{100\%}{n}\sum\left|\frac{c_i - \hat{c}_i}{c_i}\right|$

Five baseline models are compared:

1. **PLS**: Partial least-squares regression [25] on the raw spectrum.
2. **CNN-only**: The shared CNN feature extractor followed by a linear regression head, without Transformer layers.
3. **Transformer-only**: A Transformer encoder applied directly to the raw polyatomic spectrum without CNN preprocessing or mono–polyatomic decomposition.
4. **Informer**: The ProbSparse self-attention Transformer of [26], adapted for spectral regression following [18].

All models are evaluated using stratified k-fold cross-validation ($k = 5$) on the experimental dataset. Additionally, a domain-transfer experiment assesses generalisation from synthetic to experimental data without fine-tuning.

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